

Introduction to Mathematical Modeling

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1 Introduction

People make countless decisions every day of their lives. Whether it be from deciding what to have for lunch or who to vote for in the next election, we are constantly faced with choices to make based on limited information. For example, you have no idea who is working at McDonald's or how crowded it is today, but you make the decision to go there for lunch anyway since it's close and you have deals on the app. We create mental models in our minds to make decisions, qualitatively valuing some traits of a choice over another and ignoring others. These mental models work well for simple decisions with few factors to consider, but when systems become more complex and harder to conceptualize, a more advanced system is often needed. In addition, in an attempt to collaborate with others using different mental models with different values, it becomes harder to unify and make one decision that everyone can agree with. The necessity of a uniform, adaptable method of modeling has emerged from a global dependency on important decision making and the attempt to understand increasingly intricate systems around the world.

Mathematical methods of modeling have become popular due to their systematic nature and comprehensibility across different languages. Modeling is a creative process, as one problem can be approached in many different ways using different techniques. However, the principles of modeling ensure that there is little bias or oversights when solving a problem, so that another team can look over a solution and expand on it or take a different approach. This paper will be a brief overview of the process, theory, and techniques behind mathematical modeling.

Much of this will be based off of my own experience from modeling for competitions such as the HiMCM, IMMC, or M3 Challenge. These competitions give teams a multi-step modeling problem and a limited amount of time to write up a solution, following the modeling process, and then those solutions are judged against others across the world. I've learned a lot about different methods and techniques of modeling by doing these competitions.

2 The Process of Mathematical Modeling

A very important element of the modeling process is clarity. Generally, the problem that will be solved is restated so the task of the model is clear.

Each modeling approach requires a team to make simplifying assumptions about how a system operates. All of these assumptions and the justification for them are clearly stated prior to describing the model itself. For example, in trying to model the expected amount of travel time from one place to another by automobile, one could make the assumption that the rate of consumption of fuel is constant. There are many intricacies to cover to exclude extreme cases as well. We'd have to make the assumption that the vehicle is not going to break down, and use a general model of

a car instead of a truck. Every assumption changes what the model is predicting, and serves as an opportunity to extend the model to consider all possibilities. Sometimes assumptions are made because certain events are either negligible or rare enough to be negligible. After assumptions are made, terms and variables used in the model are defined, and then the technique is described and implemented.

Models can be created for multiple purposes. Sometimes, they are made just to understand a system better, which is done in many environmental scenarios. Or, models can be made to explore the effects of certain changes made to a system. This allows for the effects of scenarios to be known without performing them in real life, which can often be costly or inefficient. Another reason may be for management or control of a system, like modeling the performance of workers in an office and determining who isn't doing enough work.

After the model is created and used, it is evaluated using sensitivity analysis. Sensitivity analysis assesses how a model fares outside of situations it was built for, essentially testing how extendable it is against the assumptions that were made in its creation. Evaluation of the model also includes exploring the strengths and weaknesses of the technique used and certain improvements that can be made to it. For example, some methods of modeling may require large amounts of data before being useful, such as machine learning, while others suffer from scale, such as using a computer algorithm with high time complexity. Highlighting where your model works and where it doesn't makes it easier for others to improve upon it.

In competitions, it is often asked for teams to create a letter detailing their model's results and how it can be applied to the real world. This letter often addresses those in power to make certain changes. For example, the 2023 IMMC problem asked teams to determine the best usage for a plot of land and report the findings of their model to the business leaders that have control over that land and the power to implement what the model advised. It's important that the results of the model are clearly communicated, since when solving a real world problem you must provide a clear answer to the problem. Regardless of how many factors you took into consideration for where to go for lunch, the main tangible effect is whether you have to physically go to McDonald's or Whole Foods.

3 The Theory of Mathematical Models

This section will be focused on categorically defining different modeling types/techniques. All models deal with systems, where a system can be defined as a collection of physical objects with known attributes. A modeler can define the boundaries and aspects of a system, how objects within it interact, and how it interacts with anything outside the system (if it does). Systems that interact with their surroundings are called open, while systems that don't are closed. Variables are attributes of a system or its surroundings that have a significant effect on how the system acts. Depending on the purpose of a model, it can have different inputs and outputs. Generally, the inputs are variables that define the traits of the system and the outputs are generated by the initialized system and measured.

Sometimes, models can be based purely on data rather than taking into consideration all of the factors that go into an output. These models are often called black box models or empirical models, and are often used when the science or underlying features of a system are not fully understood.

These types of models are often used on systems in economics, such as moving average models to determine when to buy stocks.

Modeling techniques can either be deterministic or probabilistic. Probabilistic methods implement randomness if it is believed that there is an unpredictability to a system that is not captured in the variables. As a result, running a probabilistic model multiple times will result in different outcomes. Deterministic models are the opposite, there is no randomness and the system is entirely defined by the initial conditions so that the same output will be reached given the same inputs.

Models can also be categorized based on the 'time domain of a system. If time is measured over the real numbers, a system is continuous, while if time is measured over the integers a system is discrete. If the state of a system is dependent on any previous states in a model, that model is dynamic, while a static model does not depend on previous states to determine the next one.

Dynamic systems can either be lumped or distributed. These categories are based off of the function that determines future states based on the current state. This function is a function of time, and it can have other spatial variable inputs that change over time. If this function is only a function of time, the model is lumped, while if it has other inputs then the model is distributed.

Finally, models can either be analytical or numerical. These categories depend on the form of solutions to model the system. In the case of dynamic systems, which are often modelled with differential equations, there is often no closed form for solutions, so solutions are approximated numerically using the differential forms. If closed forms for solutions are available, like in most static systems, the model is analytical, and the solution allows an output to be directly obtained from an input.

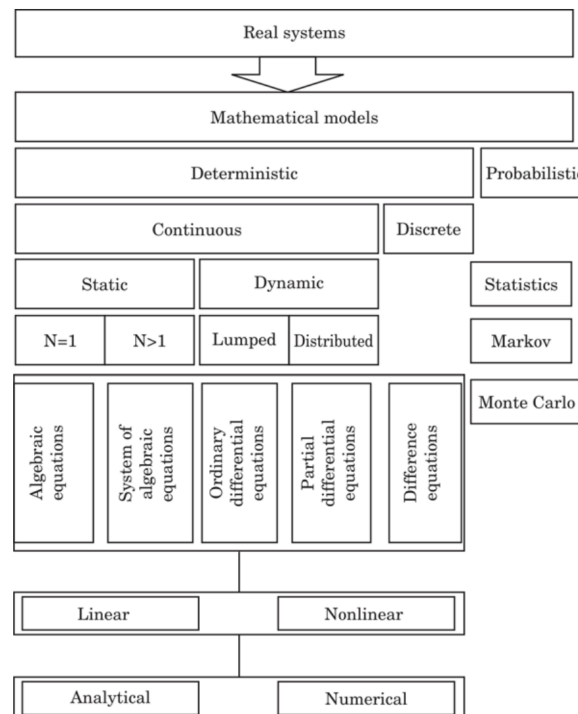


Figure 1: Chart of Types of Mathematical Models (Dr. Seiichiro Yonemur)

3.1 SQM Space Categorization

The SQM Space of Mathematical Models is another way of categorizing models. A mathematical model can be described as a triple (S, Q, M) with a system S , nature of the question Q , and mathematical statements M . This space is used to categorize real world systems in the language of mathematical models. Different systems constitute a different complexity of a model, for example we know much more about the underlying processes about electrical circuits than psychological systems. Questions can also change this, depending on how much control we want to exert over the system. There is much less certainty if we are simply trying to speculate about future states instead of implementing something sure to change future outcomes. Mathematical statements are often matched to the desired system and question. These three traits, as shown in Figure 2, can also describe a model as black box or white box. When we know more about the processes that govern a system, the system is more white box and we can use more deterministic methods for modeling. Understanding SQM space can be useful for determining what mathematical techniques to use when dealing with a certain system and question.

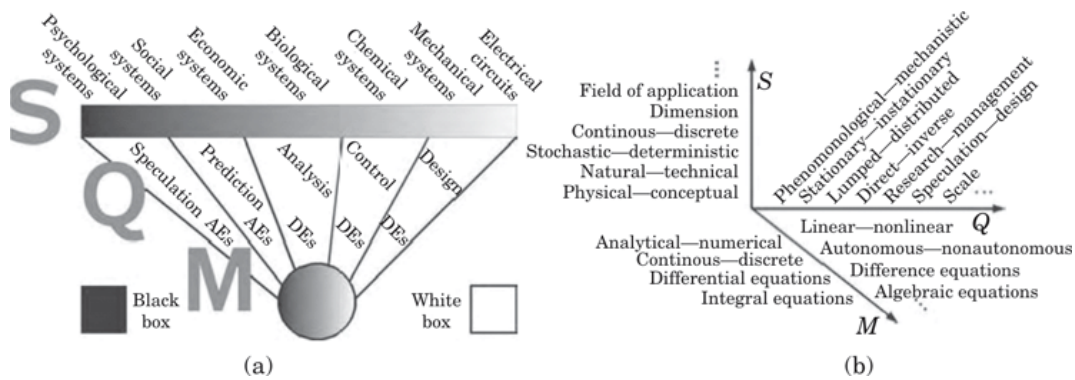


Figure 2: SQM Space, Visualized (Dr. Seiichiro Yonemur)

4 Conclusion

The process of mathematical modeling, even without looking into specific mathematical techniques, is intricate and requires careful work to follow. There is a lot more than math that goes into mathematical modeling. As Stuyvesant mathematics teacher Mr. Honner says,

“Mathematical modeling requires a lot of curiosity and creativity. Modelers must be curious about how the world works, and they must be creative in how they use math to describe it. Modelers have to be great collaborators, as no real-world problem is solved alone. And they must be great communicators, as a proposed solution will only be useful if you can convince decision makers to use it.” - Mr. Honner, 2024

References

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